

Week 8, Lecture 1 WOD: Answer Key

Newton's Method by Hand (take-home)

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. All numbers are computed in R when this file is knitted.

Plan about 8 minutes at the start of Wednesday's class to go over this, focusing on 3b and 3c. 3c leads straight into the lecture: the fixes students propose there (smaller steps, better starting values, fitting on the log scale) are what real optimizers do.

Part 1: Slope and curvature

1a.

Find $\text{NLL}'(\lambda)$ and $\text{NLL}''(\lambda)$ for $\text{NLL}(\lambda) = 6\lambda - 18 \log \lambda$. Then set $\text{NLL}'(\lambda) = 0$ and solve for the MLE, $\hat{\lambda}$.

Answer. $\text{NLL}'(\lambda) = 6 - \frac{18}{\lambda}$, $\text{NLL}''(\lambda) = \frac{18}{\lambda^2}$, and $\hat{\lambda} = 3$.

Why. $\frac{d}{d\lambda} \log \lambda = 1/\lambda$ and $\frac{d}{d\lambda} \lambda^{-1} = -\lambda^{-2}$ (the two minus signs cancel). $6 - 18/\lambda = 0$ gives $\lambda = 18/6 = 3$, the sample mean.

Watch for. A negative NLL'' (sign slip); the NLL has a minimum, so its curvature must be positive.

Part 2: Slopes without calculus

2a.

Estimate $\text{NLL}'(2)$ two ways: with the forward formula (using $\lambda = 2$ and 2.5), and with a centered version that uses $\lambda = 1.5$ and 2.5. Compare both with the exact value from 1a. Which is closer?

Answer. Forward: $(-1.4932 - (-0.4766))/0.5 = -2.0332$. Centered: $(-1.4932 - 1.7016)/1 = -3.1948$. Exact: -3 . The centered estimate is much closer.

Why. The forward difference is the slope of a chord from 2 to 2.5, which matches the curve's slope near the chord's middle, 2.25 (exact $\text{NLL}'(2.25) = -2$), not at 2. The centered chord's middle is at 2.

Watch for. Dividing the centered difference by h (0.5) instead of $2h$ (1). Sign errors from subtracting a negative number.

2b.

Curvature is the slope of the slope. Use that idea to build a finite-difference approximation to $f''(x)$ that uses only $f(x-h)$, $f(x)$, and $f(x+h)$. Then use it to estimate $\text{NLL}''(2)$ and compare with the exact value.

Answer.

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

Estimate: $(-1.4932 + 0.9532 + 1.7016)/0.25 = 4.6464$. Exact: $18/4 = 4.5$.

Why. Two forward slopes, $f'(x + h/2) \approx [f(x + h) - f(x)]/h$ and $f'(x - h/2) \approx [f(x) - f(x - h)]/h$, are a distance h apart. Their difference divided by h is the slope of the slope at x :

$$\frac{1}{h} \left[\frac{f(x + h) - f(x)}{h} - \frac{f(x) - f(x - h)}{h} \right]$$

which simplifies to the formula above.

Watch for. Dividing by h instead of h^2 (gives 2.32). Using slopes at x and $x + h$, which needs $f(x + 2h)$ and estimates the curvature at $x + h$. A student who gets the right formula by “guessing the pattern” should still be able to show where h^2 comes from.

Part 3: Newton’s method

3a.

In words: what does dividing by the curvature do to the size of the step? When will Newton’s method take big steps, and when small ones?

Answer. Strong curvature means small steps; weak curvature (a nearly straight NLL) means big steps. The sign of the slope sets the direction (downhill).

Why. A sharply curved parabola has its bottom close by; a nearly flat one has its bottom far away. Newton’s method assumes the parabola is a good stand-in for the NLL and jumps all the way to its bottom.

Watch for. “Steep slope means a big step” is only half of it: the same slope gives a small step where the curvature is large.

3b.

Start at $\lambda = 2$ and take three Newton steps, using your exact derivatives from 1a.

Answer.

Step	Guess	Slope, NLL'	Curvature, NLL''	Next guess
1	2.0000	-3.0000	4.5000	2.6667
2	2.6667	-0.7500	2.5312	2.9630
3	2.9630	-0.0750	2.0503	2.9995

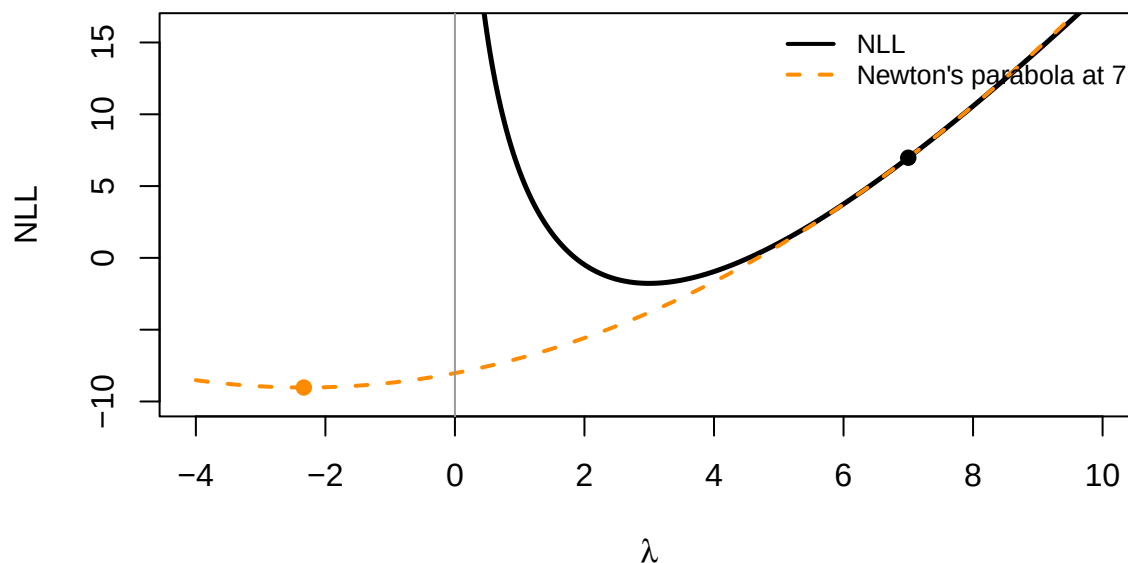
Why. The distance to the MLE goes 1, 0.33, 0.037, 0.0005: each error is roughly the square of the one before. That fast (“quadratic”) convergence near the answer is why derivative-based methods are popular.

Watch for. Rounding λ to 2.67 and carrying the rounding error forward (still converges, just more slowly). Plugging the new guess into the old curvature.

3c.

Now start at $\lambda = 7$ and take one step. What went wrong? Sketch the NLL curve and the parabola Newton’s method used at $\lambda = 7$ to explain why, and suggest one way to fix it.

Answer. $\text{NLL}'(7) = 3.429$, $\text{NLL}''(7) = 0.367$, so the next guess is $7 - 3.429/0.367 = -2.333$. A Poisson mean can’t be negative, and the NLL isn’t even defined there.



Why. Far from the MLE the NLL is nearly a straight line (small curvature), so the matching parabola is very wide and its bottom is far to the left, past zero. In general the update here simplifies to $\lambda_{\text{new}} = 2\lambda - \lambda^2/3$, which is negative for any start above 6.

Acceptable fixes (any one):

- **Shorter steps.** If a step makes the NLL worse or leaves the allowed range, halve it and try again (“backtracking”).
- **Better starting values.** Start near a sensible guess, such as the sample mean.
- **Constraints.** Use a method that respects bounds (e.g. `method = "L-BFGS-B"` with `lower = 0.001`).
- **Fit on the log scale.** Optimize over $\theta = \log \lambda$, so every value of θ gives a positive λ . Newton’s method on θ from $\lambda = 7$ gives $\lambda = 3.953, 3.106, 3.002, 3.000$: no overshoot. (This one comes up in Wednesday’s lecture.)

Watch for. Concluding Newton’s method is “broken.” It works well near the answer and badly far from it, which is why starting values matter for every optimizer.

3d.

(If you have time.) Take one Newton step from $\lambda = 2$ using your finite-difference estimates from 2a (centered) and 2b instead. How close is it to your first step in 3b?

Answer. $2 - (-3.1948)/4.6464 = 2.6876$, versus 2.6667 with exact derivatives. Close, and it would still converge.

Why. Errors in the slope and curvature only change where each step lands, not where the method stops ($\text{NLL}' = 0$). `optim()` uses a much smaller h (0.001 by default), so its estimates are closer still.

Watch for. Using the forward slope from 2a, which gives $2 + 2.033/4.646 = 2.44$; still in the right direction, but a worse step.